

Geodesic — Input Needed From the Principal

Four decisions that set the screening thresholds and currently gate every pilot verdict

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Why your answers matter now

The framework screens for a project it must first define. Every threshold in the funnel — how far is too far from water or transmission, at what load; how big a population a county can absorb — needs a denominator. We have run the full funnel (Screen → Filter → Validate → Commit) on three pilot counties against **placeholder** numbers. The pilot is complete, but **all three verdicts are conditional on the four answers below**. Until they are real, the thresholds are assumptions, not engineering.

County	Pilot verdict (draft)	Hinges on scale because...
Sweetwater, WY	Survivor — entitlement-feasible; “enter small, optioned to scale”	Absorption (labor/housing) + a 200 MW power cliff bind at full scale; the “enter small” runway shortens sharply if build-out is ~400k rather than ~50k.
Fall River, SD	Kill (confirmed) — no firm water source at scale	County pumps ~0.80 MGD total; softens to “concern” only if build-out is low-thousands.
Jones, NC	Scale-dependent — §404 wall dissolved (Sackett+SB582); water/flood/absorption migrated downstage	Could clear at ≤~15k acres; cannot at 50k. Strongest “re-open if smaller” candidate.

The four questions

1. Full build-out scale — ~50,000 people, or CF-scale (~400,000)?

WHY IT MATTERS

This is the master variable. It sets the scale-threshold ratio (project pop ÷ county pop), the water and power demand, the construction-labor and housing draw, and the land footprint — every downstream kill criterion.

HOW THE PILOT VERDICTS FLIP

At ~50k, Sweetwater works on an “enter small” basis and Jones is borderline-feasible at a reduced footprint. At ~400k, the absorption, power, and water binds tighten faster than phasing can absorb — even Sweetwater's runway shortens, and the two non-survivors are firmly dead. **This single number can move the answer for all three counties.**

YOUR ANSWER

Target full build-out population: _____ (and a realistic entry-phase size: _____)

2. Industrial anchor type?

WHY IT MATTERS

Data centers, advanced manufacturing, and logistics have wildly different power, water, and rail profiles. The derived-demand table (water/wastewater/electric/transport) is entirely mix-dependent — a heavy-power anchor changes the electric load by an order of magnitude.

HOW THE PILOT VERDICTS FLIP

Sweetwater's 200 MW regulatory cliff and Jones's grid-congestion / >100 MW integration regime both bite much harder for a data-center or heavy-manufacturing anchor than for residential-plus-light-industrial. A rail-dependent anchor also re-weights siting (Fall River's Edgemont rail is being bypassed; Jones has no in-county line).

YOUR ANSWER

Primary anchor(s) and rough on-site jobs / power intensity: _____

3. Phasing horizon — 10 years? 40 years?

WHY IT MATTERS

Phasing drives water-rights sizing and the development-agreement term length, and it determines whether “enter small, grow into the constraints” is viable or whether the project must secure full build-out supply up front.

HOW THE PILOT VERDICTS FLIP

The whole “enter small, optioned to scale” recommendation for Sweetwater depends on a phasing horizon long enough to let regional labor, housing, water-firming, and grid capacity catch up. A compressed horizon collapses that runway and pushes the at-scale binds (firm water, 200 MW, labor) onto day one.

YOUR ANSWER

Build-out horizon and rough phase plan:

4. Capital available for off-site infrastructure?

WHY IT MATTERS

This sets the “location & infrastructure” ceiling — how far the project can reach for water conveyance, transmission tie-ins, and a firm-supply solution, and therefore which constrained sites remain in play versus are priced out.

HOW THE PILOT VERDICTS FLIP

Jones is only feasible at scale via a ~\$50–120M regional Neuse-River surface-water project; Sweetwater's firm-water path runs through contracted Fontenelle storage and a customer-funded grid tap. **A low infrastructure-capital ceiling turns “solvable at a cost” into “out of reach”** and can re-rank the surviving sites.

YOUR ANSWER

Approximate off-site infrastructure budget (or ceiling): _____

Once these are answered we will (a) replace the placeholder spec, (b) re-run the affected thresholds, and (c) report which pilot verdicts change. Everything in the pilot remains a draft pending verification against official primary sources and legal counsel; these answers are the highest-value missing input.